



Montgomery County Traffic Safety Analysis (2022–Present)

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DATA 205

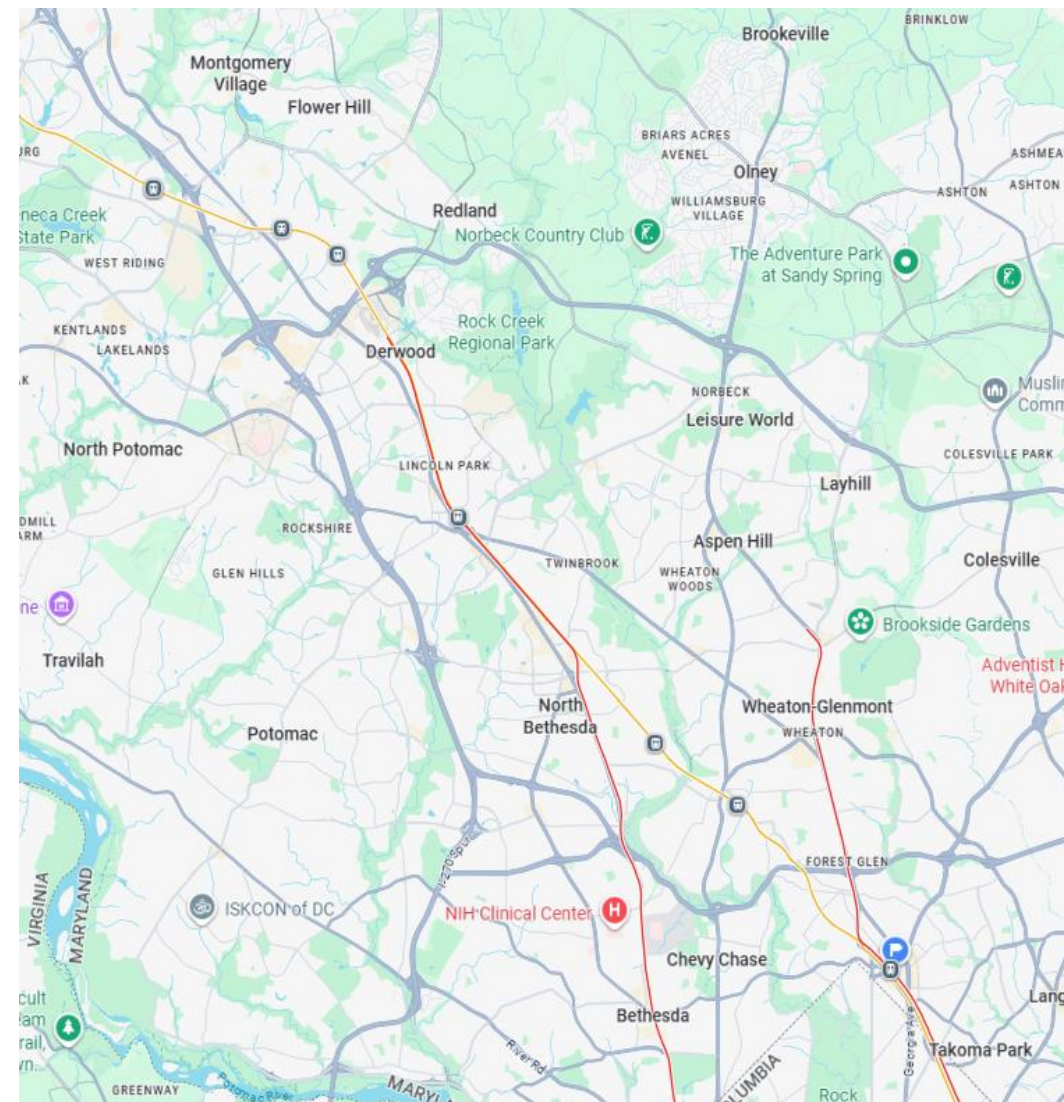
CRN: 202630

5/13/2026

PROJECT OVERVIEW

Why This Matters:

- Traffic safety affects anyone who travels, whether by driving, walking, or cycling
- Crashes and violations can lead to injuries and possibly fatalities
- Driver behavior plays a role in roadway safety
- Data analysis helps identify traffic patterns and trends
- Project combines crashes, non-motorist incidents, and violations together



DATASETS



Data Source

[DataMontgomery Open Data](#)

Original Collection System

- Maryland State Police ACRS

Reporting Agencies

- Montgomery County Police
- Gaithersburg Police
- Rockville Police
- Maryland-National Capital Park Police

Important Note

- Data may contain preliminary or unverified information

Crash Reporting: Drivers Data

Includes crash severity, weather conditions, timestamps, light levels, if at fault and location information.

- **Total Records:** 7,000+
- **Main Variables:** Injury Severity, Weather, Road Conditions, Hour, Date of Stop, Coordinates

Traffic Violations Data

Includes demographic information, citation descriptions, and stop locations.

- **Total Records:** 296,000+
- **Main Variables:** Description, Gender, Race, Date of Stop, Coordinates

Crash Reporting: Non-Motorists Data

Includes, timestamps, injury severity, if at fault, and if pedestrians and cyclists

- **Total Records:** 2,700+
- **Main Variables:** At Fault, Injury Severity, Month, Date of Stop

TOOLS USED & CLEANING PROCESS

Tools:

ggplot2, plotly, leaflet, Tableau Public, CSV datasets

Cleaning Steps

- Removed records missing latitude/longitude
- Filtered data from 2022–present
- Extracted month, year, and hour variables
- Removed unknown values when necessary
- Combined duplicate traffic violations and crashes using their respective identifiers
- Merged repeated citation receivers' descriptions
- Merged values with differing grammar

```
### Removing columns not in use
```

```
crash_data$'Person ID' <- NULL
crash_data$'Report Number' <- NULL
crash_data$'Agency Name' <- NULL
crash_data$'Route Type' <- NULL
crash_data$'Cross-Street Name' <- NULL
crash_data$'Off-Road Description' <- NULL
crash_data$'Municipality' <- NULL
crash_data$'Non-Motorist Substance Abuse' <- NULL
crash_data$'Vehicle ID' <- NULL
crash_data$'Location' <- NULL
crash_data$'Parked Vehicle' <- NULL
crash_data$'Vehicle Going Dir' <- NULL
crash_data$'Vehicle Damage Extent' <- NULL
crash_data$'Vehicle First Impact Location' <- NULL
```

```
'''{r}
crash_data
```

Clean Weather

```
crash_data$weather <- ifelse(crash_data$weather %in% c("RAINING", "Rain", "Freezing Rain Or Freezing Drizzle"), "Rain",
  ifelse(crash_data$weather %in% c("snow", "SNOW", "Blowing Snow", "SLEET", "WNTNRY MIX"), "Snow",
    ifelse(crash_data$weather %in% c("Severe Crosswinds", "SEVERE WINDS"), "Windy",
      ifelse(crash_data$weather %in% c("BLOWING SAND, SOIL, DIRT", "cloudy", "Fog, Smog, Smoke", "FOGGY"),
        ifelse(crash_data$weather %in% c("clear", "CLEAR"), "Clear", crash_data$weather))))))
```

```
crash_data <- filter(crash_data,
  !is.na(Weather), Weather != "",
  !Weather %in% c("Other", "Unknown", "UNKNOWN", "OTHER", "N/A"))
```

```
unique(crash_data$'weather')
```

Clean Light Level

```
crash_data$Light <- gsub(" ", "", crash_data$Light)
```

```
crash_data$Light <- ifelse(crash_data$Light %in% c("DARK--UNKNOWNLIGHTING", "Dark-UnknownLighting", "Dark-NotLighted", "Dark-Lighted", "DARKLIGHTSON", "DUSK", "Dusk", "Dawn", "DAWN"),
```

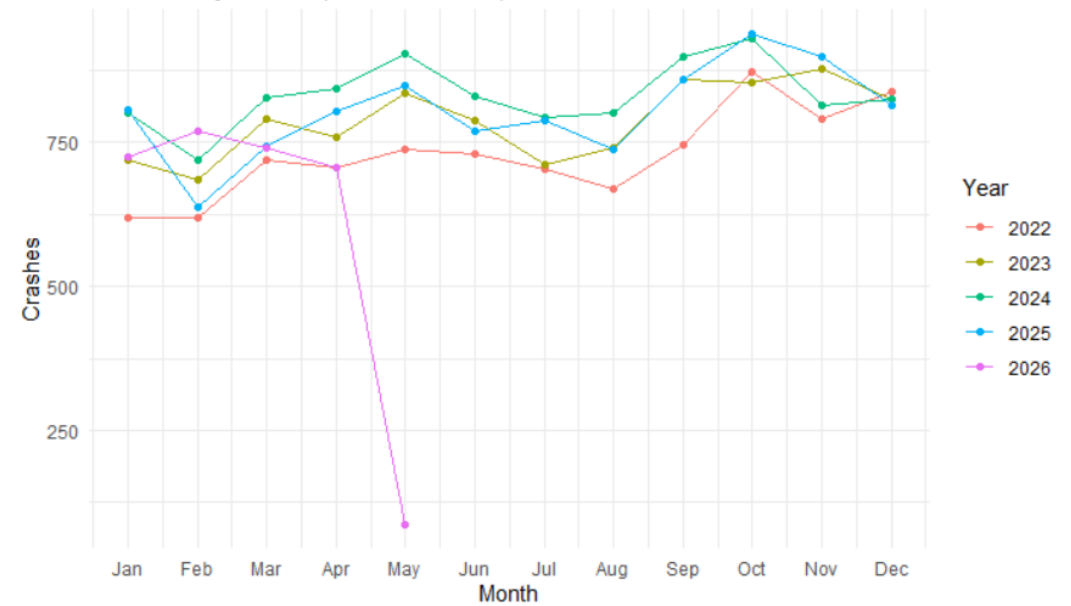
```
crash_data <- filter(crash_data,
                     !is.na(Light), Light != "")
```

CRASH TRENDS OVER TIME

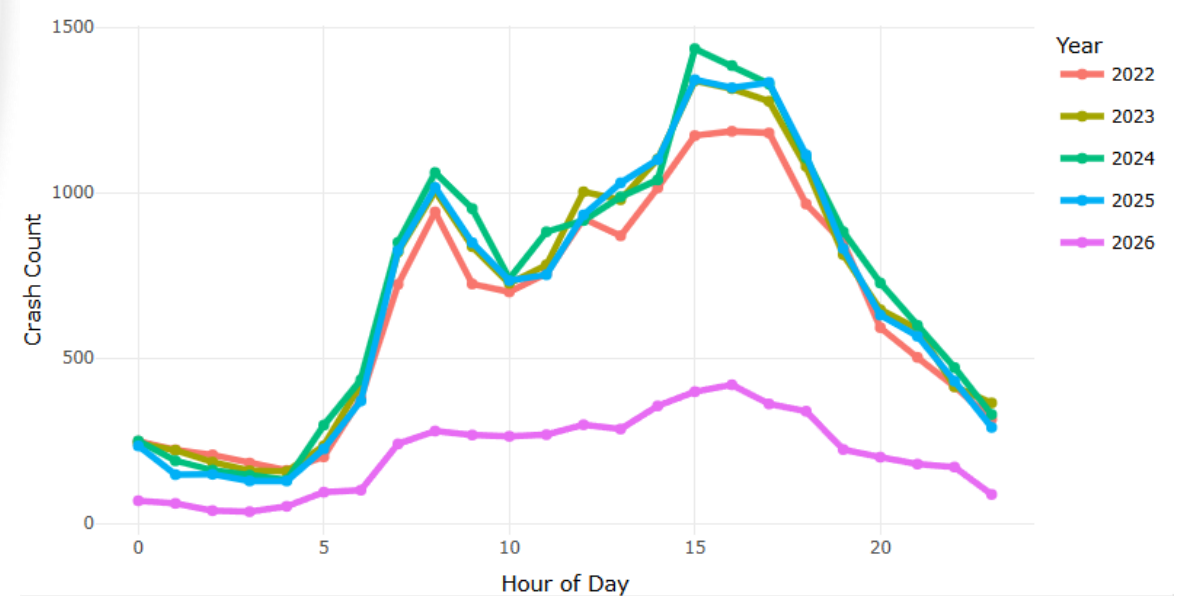
Key Findings

- Crash activity showed recurring seasonal patterns
- September–November consistently showed increased activity
- Crash activity increased during busier roadway periods
- Seasonal commuting and reduced daylight may contribute to trends

Crashes by Month (2022–Present)



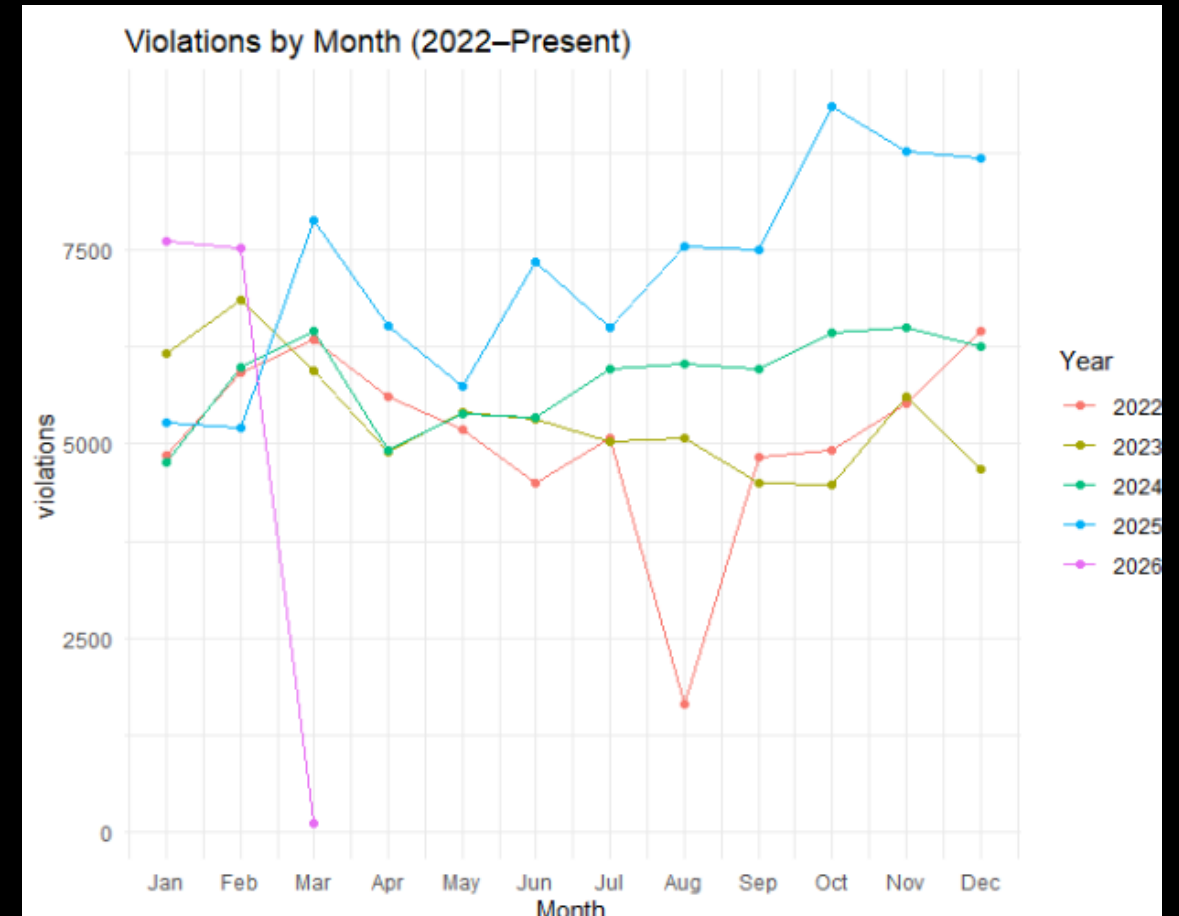
Crashes by Hour



TRAFFIC VIOLATION ANALYSIS

Observations

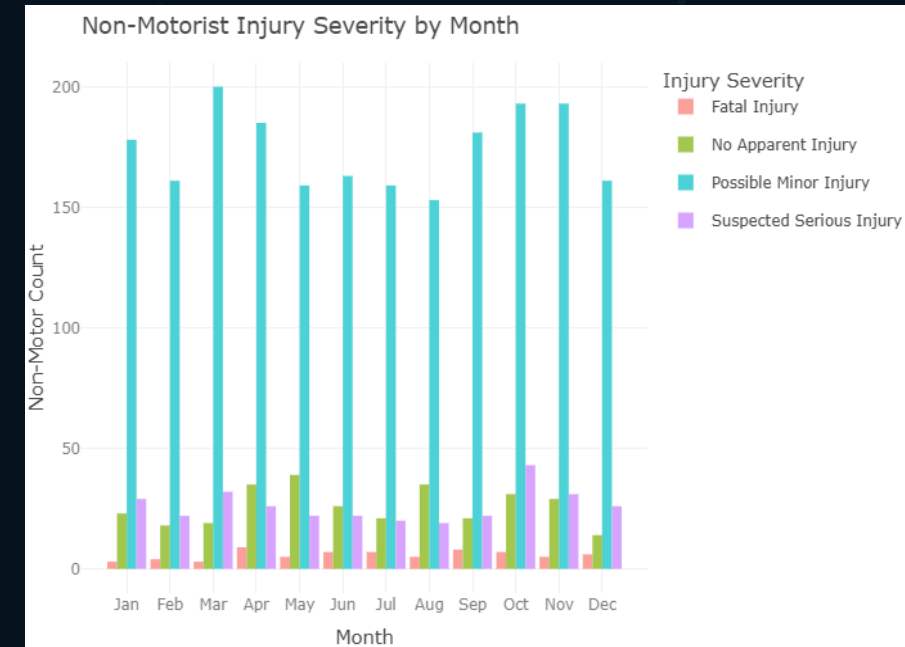
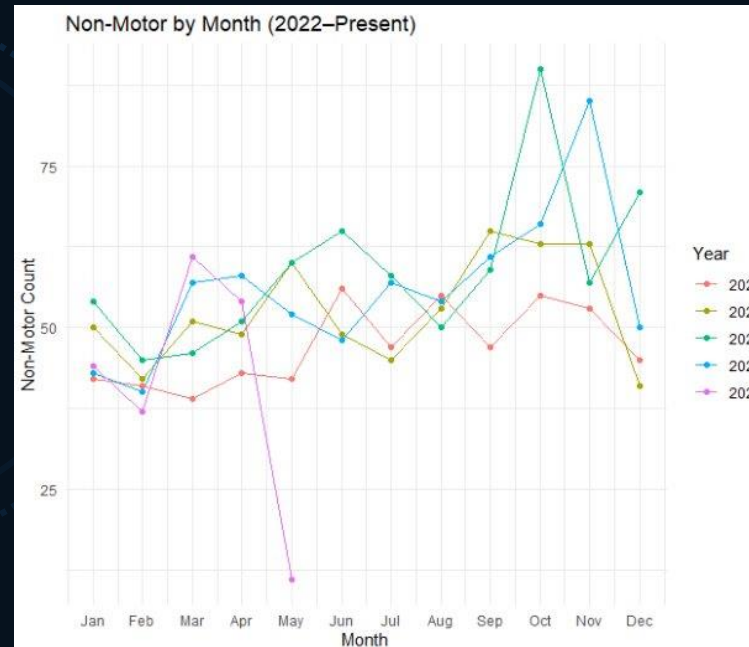
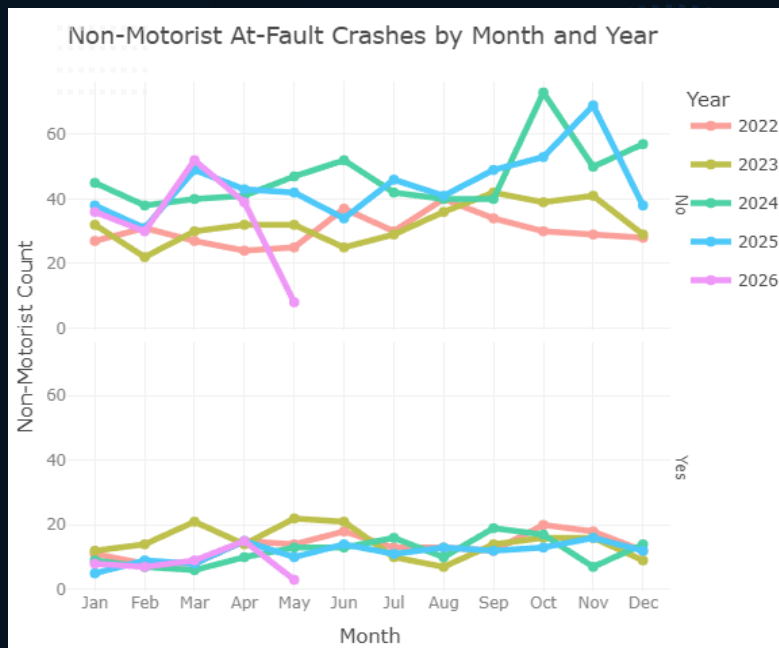
- Violation activity followed patterns similar to crash activity
- September – November showed increased activity
- Higher activity occurred during busy traffic periods
- Trends may reflect increased traffic density and driver interaction



NON-MOTORIST TRENDS OVER TIME

Observations

- Non-motorist activity followed patterns similar to crash activity
- Non-motorists were sometimes listed at fault
- Injury severity varied across incidents
- Reduced physical protection may increase injury severity

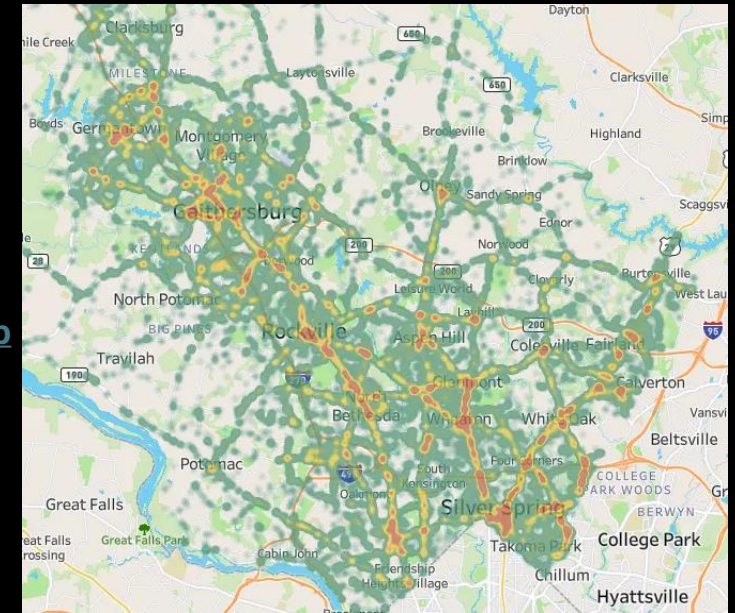


GIS HEAT MAP & HOTSPOTS

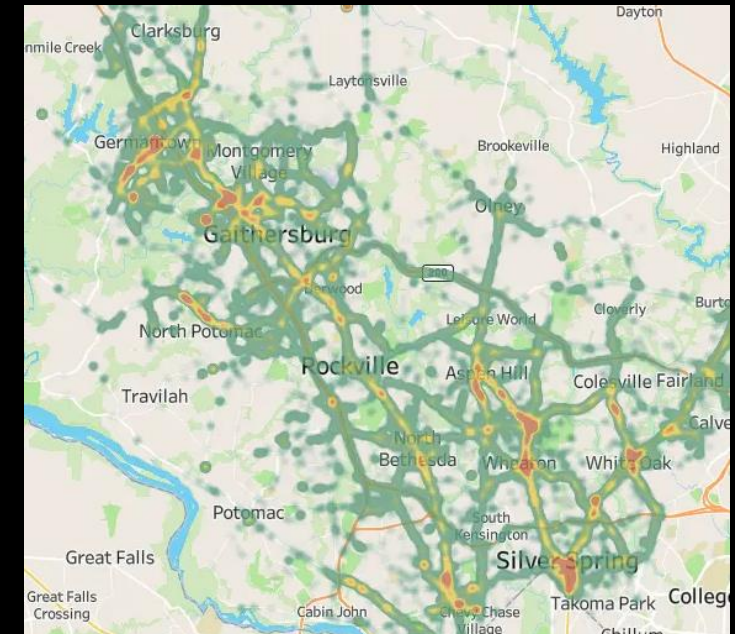
Observations

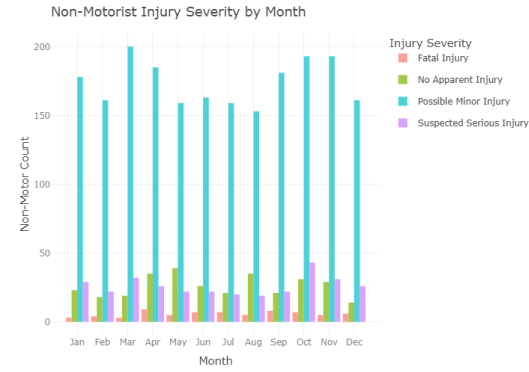
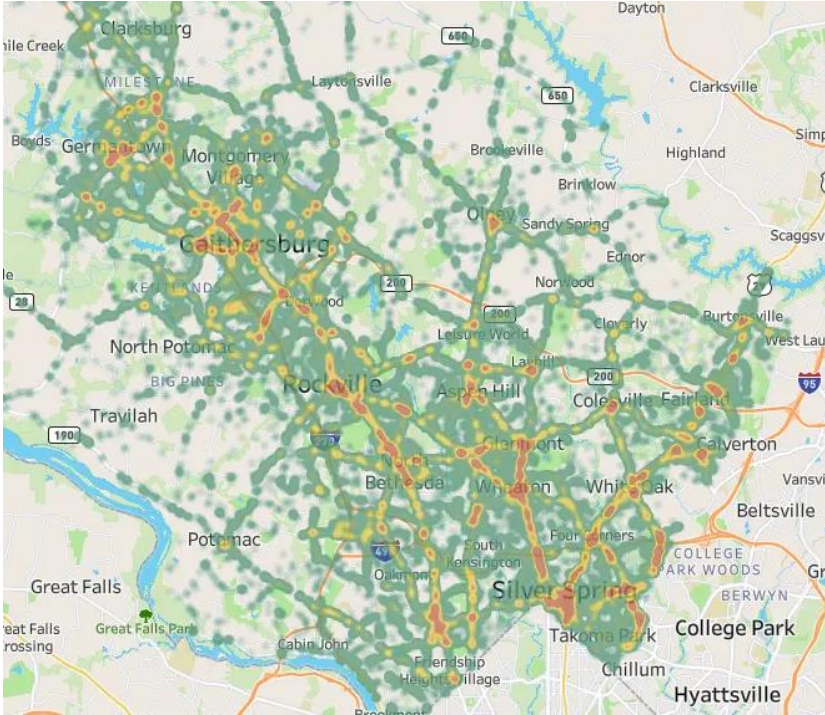
- Crash and violation hotspots appeared near intersections
- Highways showed lower density than expected
- Intersections may experience greater activity due to turning, congestion, and pedestrian interaction
- Spatial patterns were consistent across both heat maps

Crash Data
Geospace Map



Violations
Geospace Map

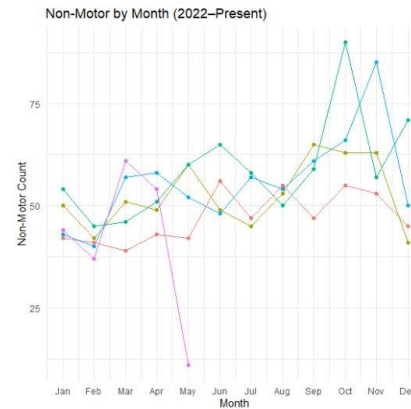




CONCLUSIONS

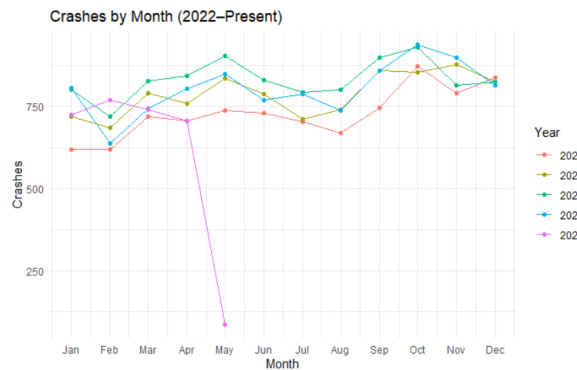
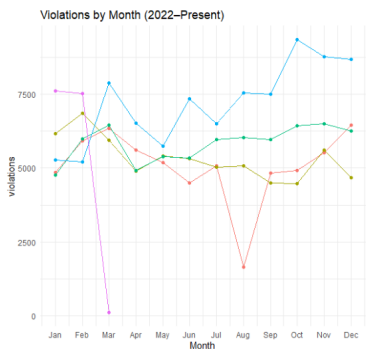
Key Findings

- Around October showed the most increased activity
- Violations followed patterns similar to crashes
- Non-motorists faced more injury involvement
- Intersections showed higher crash density



Value Provided

- Combining datasets revealed broader traffic patterns
- GIS analysis identified high-risk areas
- Visualizations improved trend interpretation
- Findings highlight roadway safety concerns



References & Acknowledgements

Acknowledgements

- Thank you Professor Perni for an informative semester along with my classmates forgiving feedback and seeing through the eyes of the audience during this project.
- Thank you to Professor Saidi for showing me the basics of R back in summer 2025 and introducing me to Tableau for GIS mapping.

• Tools & Libraries

- Data Montgomery
- Maryland State Police ACRS
- R Studio
- Tableau Public
- GitHub
- Course resources
- ggplot2, plotly, leaflet, CSV datasets

Datasets

Data Montgomery Open Data: <https://data.montgomerycountymd.gov/>

Crash Reporting - Drivers Data: https://data.montgomerycountymd.gov/Public-Safety/Crash-Reporting-Drivers-Data/mmzv-x632/about_data

Traffic Violations Data: https://data.montgomerycountymd.gov/Public-Safety/Traffic-Violations/4mse-ku6q/about_data

Crash Reporting - Non-Motorists Data: https://data.montgomerycountymd.gov/Public-Safety/Crash-Reporting-Non-Motorists-Data/n7fk-dce5/about_data